



COMMENTS ON PROFESSIONAL ETHICS

FOR

**Dr. Roy Hartfield's
AERO 4510 STUDENTS**

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ENGINEERING

(13) PRACTICE OF ENGINEERING.

a. Any professional service or creative work, the adequate performance of which requires engineering education, training, and experience in the application of special knowledge of the mathematical, physical, and engineering sciences to such services or creative work that includes any one or more of the following:

1. Consultation, investigation, evaluation, planning, design and design coordination, or commissioning of engineering works, products, and systems.
2. Planning the use of land, air, or water.
3. Performing engineering surveys and studies.
4. The review of construction for the purpose of monitoring compliance with drawings and specifications.

•b. Any service or work described in paragraph a., either public or private, that is made in connection with any utilities, structures, buildings, machines, equipment, processes, work systems, projects, communications systems, transportation systems, industrial or consumer products, or equipment of a control system; or is of a mechanical, electrical, hydraulic, pneumatic, chemical, environmental, or thermal nature, insofar as the service or work involves safeguarding health, life, safety, welfare, and property. The term includes other professional services as may be necessary to the planning, progress, and completion of any engineering services.



ENGINEERING

d. The practice of engineering includes the offering of expert opinion in any legal proceeding in Alabama regarding work legally required to be performed under an Alabama engineer's license number or seal, which opinion may be given by an engineer licensed in any jurisdiction.

Notwithstanding any other provision of this chapter, in qualifying a witness to offer expert testimony on the practice of engineering, the court shall consider as evidence of his or her expertise whether the proposed witness holds a valid Alabama license for the practice of engineering; provided, however, the qualification by the court shall not be withheld from an otherwise qualified witness solely on the basis of the failure of the proposed witness to hold a valid Alabama license or license from another jurisdiction.

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(Acts 1961, Ex. Sess., No. 79, p. 1976, §2; Acts 1967, No. 739, p. 1576, §1; Acts 1997, No. 97-683, p. 1332, §1; Act 2002-514, p. 1323, §1; Act 2007-365, p. 715, §1; Act 2014-375, p. 1390, §1; Act 2018-550, §1.)



**“Always do right. This will gratify some people and
astonish the rest.”** Mark Twain



WHAT IS ETHICS?

“[Ethics is][t]he science of moral values and duties.”

Webster’s New Collegiate Dictionary, 1960.

Ethics is the science that attempts to classify what people do (“conduct”) as “good” or “bad,” and which explores the reasons why conduct is considered to be “good” or “bad.” It pertains to *voluntary*, as opposed to *involuntary* action. Thus, it depends on a rational thought process.



- **Is “ethics” a natural, pre-programmed characteristic of human beings, or is ethical behavior learned?**

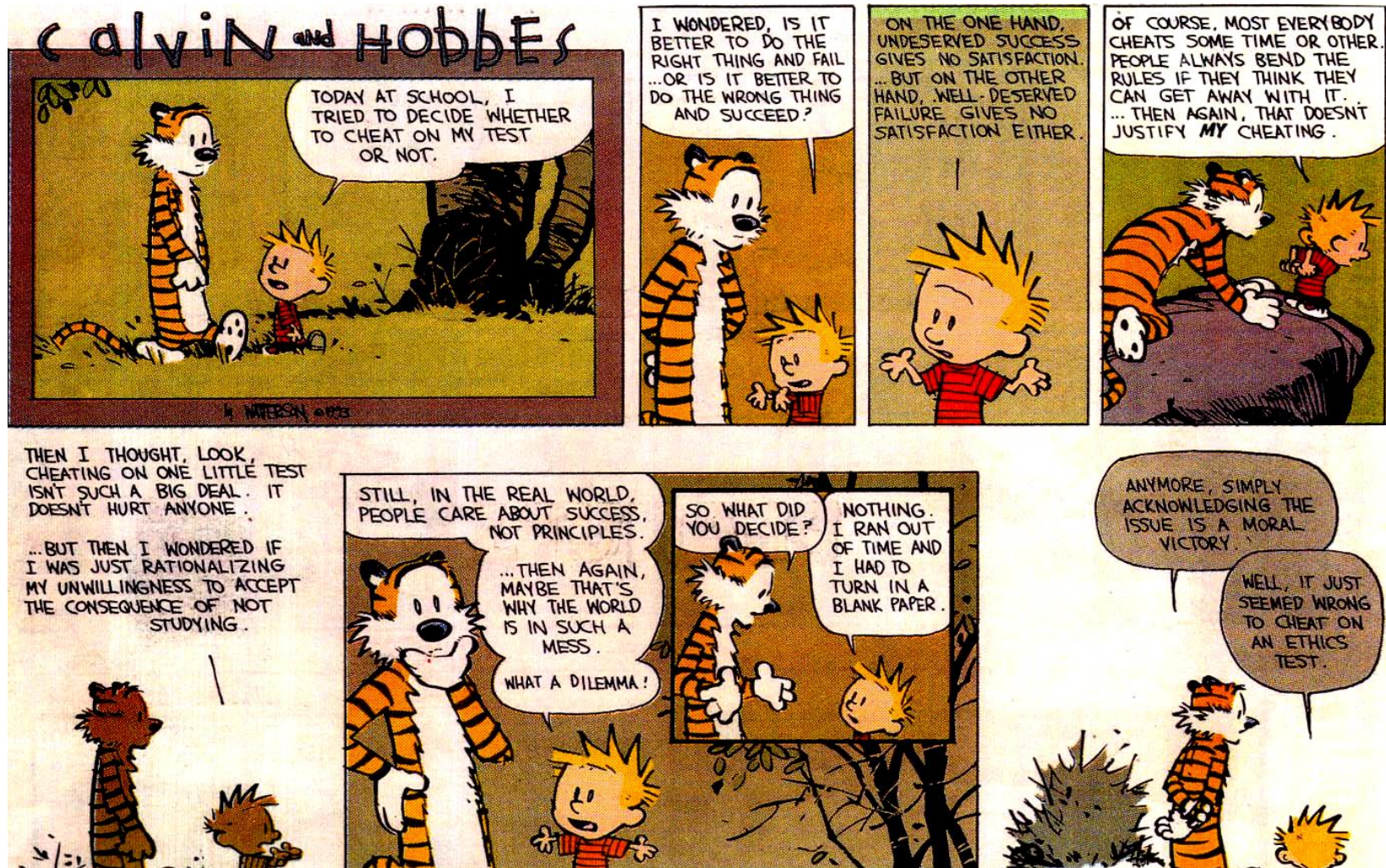
One Answer: Most humans are concerned about what is good and what is bad, some more so than others. It does not appear that we are any more born to be “ethical” than we are born to be leaders.

- **Ethics and Religious Convictions**

You do not have to be religious to be ethical. However, most religions teach ethical behavior.

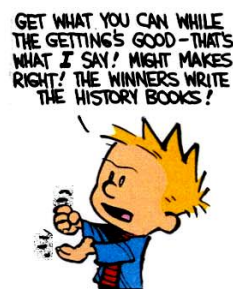
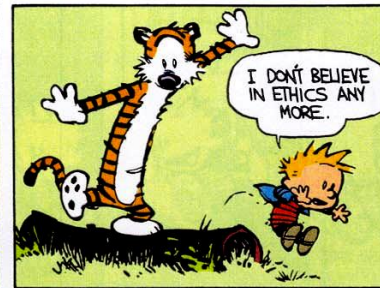
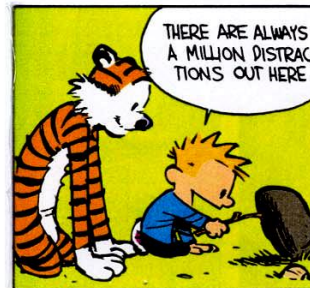


• CALVIN ON ETHICS





• MORE OF CALVIN ON ETHICS





PERSONAL SYSTEMS OF ETHICS

Do we need personal systems of ethics?

Most people would say we do. “Society” depends on the existence of rules. Some of these are “laws,” rules enforceable directly by an instrument of society (kings, courts, ...) through the use of penalties for violating the rules.

- **Personal Ethics** is a set of rules that are self-enforced.
- **Professional Ethics, on the other hand, is a set of rules, usually contained in a Code of Ethics for the particular profession.** Certain rules of a profession are enforceable by the profession and usually by the state in which the professional practices.



WHAT IS A PROFESSION?

- DEFINITIONS

"A profession is a vocation, a calling, or an employment requiring special knowledge, skill, education ... predominantly mental or intellectual rather than physical."

A vocation, or employment, the carrying out of the duties and responsibilities of which may have a significant effect on the public is generally classified as a profession.

Because of the potential for harm from the projects, or other end products of the work, of those who "practice," professions are regulated by the state.



PROFESSIONAL ETHICS

- The “Learned Professions”: Theology, Law, Medicine
- Modern Professions: Those requiring licensure by the state in which the profession is practiced (except for the clergy).

**ALTHOUGH IN SOME RESPECTS VERY OLD,
ENGINEERING IS A MODERN PROFESSION**



PROFESSIONAL ETHICS

- "Ethics are principles of conduct governing an individual or group."

Webster's Ninth New Collegiate Dictionary, Merriam-Webster, Inc.
Springfield, Massachusetts, 1983.

- "Honesty, justice and courtesy form the moral philosophy which, associated among mankind, constitute the foundation of ethics." from the forward of the 1947 ECPD (Engineering Council for Professional Development) *CANONS OF ETHICS*.



PERSONAL ETHICS vs. ETHICS FOR ENGINEERS

- As noted above, we develop personal concepts of what is right and wrong.
- In many cases, professional ethics are more restrictive than the personal ethics of the parties.
- Engineers must not only refrain from unethical acts, but they must also **actively avoid circumstances** that would raise a question of impropriety.



HISTORY OF PROFESSIONAL ETHICS FOR ENGINEERING.

2500 B.C. -- KING HAMMURABI, MILITARY "ENGINEERING"

1750 A.D. -- JOHN SMEATON FIRST USED THE TERM "CIVIL ENGINEERING."

1818 -- INSTITUTION OF CIVIL ENGINEERS FORMED IN GREAT BRITAIN.

1852 - AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE) FOUNDED

1906 - AMERICAN INSTITUTE OF ELECTRICAL ENGINEERS, ADDRESS OF DR. WHEELER ON ENGINEERING HONOR INSPIRED MEMBERS TO DRAFT A **CODE OF ETHICS. ADOPTION WAS DELAYED AND PROPOSAL DIED BECAUSE OF LACK OF INTEREST.**



1906 -- FIRST ENGINEERING CODE OF ETHICS ADOPTED BY THE CIVIL ENGINEERS OF ENGLAND.

1908 -- AMERICAN INSTITUTE OF CHEMICAL ENGINEERS WAS FOUNDED AND ADOPTED A CODE OF ETHICS.

1911 -- AMERICAN SOCIETY OF CONSULTING ENGINEERS ADOPTED A CODE.

DURING THE EARLY 1900'S THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS SOUGHT TO DEVELOP A CODE FOR ALL ENGINEERING SOCIETIES. ASME ADOPTED THE CODE. OTHER SOCIETIES DID NOT.



1914 -- AMERICAN SOCIETY OF CIVIL ENGINEERS' "SHORT CODE" BASED ON THE ASME CODE.

1932 -- THE ENGINEERS COUNCIL FOR PROFESSIONAL DEVELOPMENT (ECPD) WAS FORMED. (ACCREDITATION, ETHICS, TECHNIQUES FOR VOCATIONAL GUIDANCE, MEANS FOR TRANSITION FROM COLLEGE TO WORK).

1934 -- NATIONAL SOCIETY OF PROFESSIONAL ENGINEERS FORMED. (SOCIAL, ECONOMIC, POLITICAL, PROFESSIONAL INTERESTS OF ENGINEERS)

1941 -- ECPD UNDERTAKES DEVELOPMENT OF ETHICAL STANDARDS FOR ALL ENGINEERS.

1958 -- ECPD CANONS OF ETHICS ADOPTED.

1963 -- ECPD CANONS OF ETHICS REVISED.



1976 -- "CODE OF ETHICS FOR ENGINEERS" ADOPTED BY ECPD.

14 PARTICIPATING SOCIETIES. 8 ENDORSED THE FUNDAMENTAL PRINCIPLES; 6 THE FUNDAMENTAL CANONS; 1 THE GUIDELINES.

**FUNDAMENTAL PRINCIPLES -- 4 ELEMENTS
FUNDAMENTAL CANONS -- 7 ARTICLES
SUGGESTED GUIDELINES -- 56 ELEMENTS**



EXAMPLES OF CODES

Alabama Law (CODE OF ALABAMA 2022, TITLE 34, SECTION 34-11-11)

NSPE

AIAA

Boeing, Lockheed Martin, Northrop Grumman, ...



LEGAL ENFORCEMENT OF THE CODE OF PROFESSIONAL CONDUCT

CODE OF ALABAMA 1975, TITLE 34, SECTION 34-11-11

The State Board of Registration for Engineers and Land Surveyors has the ". . . power to censure, place on probation and/or fine any registrant or certificated engineer-in-training or corporation, partnership or firm holding a certificate of authorization and to suspend, refuse to renew and/or revoke the certificate of any registrant or certificated engineer-in-training or the certificate of authorization of a corporation, partnership or firm found guilty of: . . . a violation of the rules of professional conduct (code of ethics) prescribed by the board or any amendment thereof . . ."

See www.bels.alabama.gov/ **Re Professional Registration and Enforcement**
www.bels.alabama.gov/law.htm

NSPE

CODE OF ETHICS FOR ENGINEERS

FUNDAMENTAL CANONS

ENGINEERS, IN THE FULFILLMENT OF THEIR PROFESSIONAL DUTIES, SHALL:

- 1. HOLD PARAMOUNT THE SAFETY, HEALTH, AND WELFARE OF THE PUBLIC IN THE PERFORMANCE OF THEIR PROFESSIONAL DUTIES.**
- 2. PERFORM SERVICES ONLY IN THEIR AREAS OF COMPETENCE.**
- 3. ISSUE PUBLIC STATEMENTS ONLY IN AN OBJECTIVE AND TRUTHFUL MANNER.**
- 4. ACT IN PROFESSIONAL MATTERS FOR EACH EMPLOYER OR CLIENT AS FAITHFUL AGENTS OR TRUSTEES.**
- 5. AVOID DECEPTIVE ACTS.**
- 6. CONDUCT THEMSELVES HONORABLY, RESPONSIBLY, ETHICALLY, AND LAWFULLY SO AS TO ENHANCE THE HONOR, REPUTATION, AND USEFULNESS OF THE PROFESSION.**



AIAA CODE OF ETHICS

<https://www.aiaa.org/codeofethics/>

- 1. Members of the AIAA uphold the highest standards of ethical conduct and hereby agree to:**
- 2. Hold paramount the safety, health, and welfare of the public in the performance of their duties;**
- 3. Promote the lawful and ethical interests of AIAA and the aerospace profession;**
- 4. Reject bribery, fraud, and corruption in all their forms;**
- 5. Properly credit the contributions of others, accept and offer honest and constructive criticism of technical work; and acknowledge and correct errors;**
- 6. Avoid harming others, their property, their reputations or their employment through false or malicious statements or through unlawful or otherwise wrongful acts;**
- 7. Issue statements or present information in an objective and truthful manner, based on available data;**



- 8. Undertake only those tasks for which we are qualified by training or experience, or for which we can reasonably become qualified with proper preparation, education, and training;**
- 9. Maintain and improve our technical and professional competencies throughout our careers and provide opportunities for the professional development of those engineers under our supervision;**
- 10. Treat fairly and respectfully all colleagues and co-workers, recognizing their unique contributions and capabilities.**

Adopted 9 May 2013.



See Guidelines to Practice under the AIAA Code of Ethics for additional detail.

•All allegations of ethical misconduct shall be addressed in writing to the Chair of the AIAA Ethics Committee. The complaint shall specify the section of the AIAA Code of Ethics that is alleged to have been violated. It shall state the facts of the alleged violation and provide any available evidence in support of the allegations. These may be submitted either by email to Emily Springer, AIAA Ethics Committee Liaison: emilys@aiaa.org, or submitted by mail to:

•AIAA Ethics Committee

American Institute of Aeronautics and Astronautics

12700 Sunrise Valley Drive, Suite 200

Reston, Virginia 20191-5807

Attn: Emily Springer, AIAA Ethics Committee Liaison

Email: emilys@aiaa.org



- **Boeing, Lockheed Martin, Northrop Grumman**

Northrop Grumman – Dr. Ron Sugar part of address to the Naval Academy 2007

*“What is DII? It stands for **Defense Industry Initiative**, and it represents another important way that corporate ethics can be promoted and advanced – by self-policing industry-wide... It was conceived within a maelstrom of scandals, allegations and improprieties that spawned the Packard Commission study. **In 1986, that commission reported that the American public had a lack of confidence in our industry.** It concluded that defense companies could benefit from a greater emphasis on corporate self-governance. Soon thereafter, the CEOs of that time got together to draft the principles that would become the Defense Industry Initiative.*



- **Boeing, Lockheed Martin, Northrop Grumman**

Northrop Grumman (continued)

Far from window dressing, the DII has real ballast to it. Each DII signatory pledges to uphold six fundamental principles:

“They must adhere to written codes of conduct;

They must train employees in those codes;

They must encourage the internal reporting of code violations;

They must practice self-governance;

They must share ‘best practices’ with other firms.

And they must be accountable to the public.”



EXAMPLES AND CASES

Space Shuttle Challenger Disaster

January 28, 1986

DC-10 Cargo Door Latch Failure

June 1972 – Until it was fixed!



EXAMPLES AND CASES

Space Shuttle Challenger Disaster

The Space Shuttle Challenger, with teacher Christa McAuliffe aboard, exploded in flames on live television on January 28, 1986. Because of the intense public interest in the explosion and the fiery death of the astronauts, the Challenger case has been fully publicized.

The direct cause of the Challenger explosion was technical - faulty O-rings. But the Challenger also presented a case study in organizational communication and ethics, including the ethics of organizational structure and culture as it promotes or discourages necessary communication, the **ethics of whistle blowing**.



EXAMPLES AND CASES

DC-10 Cargo Door Latch Failure





EXAMPLES AND CASES

http://en.wikipedia.org/wiki/McDonnell_Douglas_DC-10

DC-10 Cargo door problem

A problem with the outward-opening cargo door was first identified on June 12, 1972, when American Airlines Flight 96 lost its aft cargo door after takeoff from Detroit, Michigan. Fortunately, the crew was able to perform an emergency landing with some of the aircraft's control surfaces disabled. Aileron controls and differential thrust of the wing engines were used to control the DC-10. Before Flight 96 took off, an airport employee had forced the door shut, weakening the locking pin and causing the door to blow out as the airliner reached altitude.^[27]

The DC-10 was designed with cargo doors that opened outward instead of inward-opening "plug-type" doors. Using outward-opening doors allowed the cargo area to be completely filled since the door was not occupying usable space. To secure the door against the outward force caused by the pressurization of the fuselage, outward-opening doors rely on a heavy locking mechanism. In the event of a complete door lock malfunction, there was potential for explosive decompression.



EXAMPLES AND CASES

http://en.wikipedia.org/wiki/McDonnell_Douglas_DC-10

In 2006, [FedEx](#) became the first U.S. carrier to equip its aircraft with an anti-missile defense system. The gray oval [Northrop Grumman Guardian](#) pod can be seen on the belly of this FedEx MD-10 between and just aft of the main landing gear.

Although many carriers voluntarily modified the cargo doors and re-trained their ground crews, there was not yet a mandatory rework of the system. Severe problems still persisted with the aircraft's cargo doors, and two years after the American Airlines incident, an almost identical cargo door blow-out caused [Turkish Airlines Flight 981](#) to crash into a forest near the town of [Ermenonville](#) shortly after leaving Paris on March 3, 1974, killing 346 people in one of the deadliest air crashes of the twentieth century. Circumstances of this crash were similar to the previous accident. However, a modified seating configuration on the Turkish aircraft exacerbated the effects of decompression which caused the floor of the aircraft to collapse into the cargo bay. As part of the DC-10 design, vents were not adequate to allow the pressure between the cargo and passenger compartments to equalize. Control cables running through the floor of the plane were severed when the floor collapsed and this rendered the aircraft uncontrollable. Following this crash, all DC-10s underwent mandatory door modifications.



DC10 Cargo Door (cont.)

June 27, 1972 -- Daniel Applegate, Director of Product Engineering for Convair, the fuselage contractor, wrote a memo to his supervisors detailing potential problems of cargo door. The problem was first recognized in August 1969. The same thing had also happened in a ground test in 1970.

Recognized design flaws – door, floor, latch.



DC10 Cargo Door (cont.)

After the Detroit near-disaster, the NTSB (National Transportation Safety Board) investigation revealed several problems and recommended immediate design changes. **The FAA did not follow the NTSB's recommendations.** FAA director John Shaffer and Douglas President Jackson McGowan reached a gentleman's agreement to voluntarily fix problem, but no further official action was taken.

In July 1972, three inspectors at Long Beach plant certified that Ship 29 had been modified (but it was not). **Two years later, after leaving Paris, its cargo door blew off at 13,000 feet, killing 346 people.**



DC10 Cargo Door (cont.)

Why did the Door Blow-Off? Who was responsible?

Questions existed as to whether same design principles question as to whether same design principles apply to wide apply to wide-body aircraft body aircraft.

However, in designing the airframe, McDonnell Douglas Douglas chose to make DC-10 much like the older DC-8 and DC-9, which had been very successful and safe aircraft.

Reasons:

- 1. Use of design methods much like that of previous aircraft sped up the design process.**
- 2. It fit with “an oft an oft-stated company policy of stated of technological caution.”**



QUESTIONS

- **What should you, as a junior engineer, do if you know of such a problem with an airplane your company manufactures?**
- **Who should report these types of things, how and why?**



CODES OF ETHICS

CANNOT GUARANTEE PROPER BEHAVIOR



SUMMARY

- **Professional ethics differs from personal ethics.**
- **Professionals are held to higher ethical standards than laypersons.**
- **Engineering is a profession.**
- **Engineers who are registered may lose their right to practice engineering if they violate their state's code of professional conduct.**
- **Most engineering organizations have their respective codes of ethics. The AIAA does.**
- **A good model for a code of ethics is the NSPE Code.**
- **Even engineers who are not registered should follow appropriate rules of professional responsibility.**
- **Many events that have claimed the lives of persons and the destruction of property could have been avoided if the engineers involved had acted ethically.**
- **Know the rules of professional responsibility that apply you and follow them rigorously.**



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